

**1. What do you mean by transport plus concept? Which agency in Nepal is adopting this concept? Why this concept is important in Nepalese mountainous and hilly roads? Write down pros and cons of this concept.**

Answer:

Transport plus concept is a rural road development approach that pairs road construction with supplementary community programs, local capacity building, and livelihood support. It combines building physical infrastructure with social and economic upliftment for nearby residents. Instead of only constructing roads or bridges, the project also promotes services and activities that improve the quality of life and support local development.

The Department of Local Infrastructure (DoLI) is one of the major agencies adopting the Transport Plus concept, particularly in rural road development projects. This approach has also been promoted through projects supported by development partners such as the Swiss Agency for Development and Cooperation (SDC) and implemented in various rural road programs in Nepal.

The topography of Nepal is characterized by an extreme range of elevations and a dramatic geographical transition from the high-altitude Himalayas to the flat Terai plains. Simply constructing roads is often insufficient in such rugged and complex topography as it requires locales support. The Transport Plus concept is important because it:

- i. Transport plus concept enhances rural accessibility to markets, schools, and health facilities.
- ii. It creates local employment during road construction and maintenance.
- iii. It also promotes environmental conservation through bioengineering and slope stabilization.
- iv. The concept reduces landslides and soil erosion by integrating proper drainage and vegetation.
- v. It also improves road safety through awareness programs and safety measures.
- vi. It supports local economic development, tourism, and agriculture.
- vii. This approach encourages community participation, resulting in better ownership and maintenance.
- viii. It also makes road investments more sustainable and beneficial to local people.

Pros of transport plus concept:

- i. This concept generates employment and income for local communities.
- ii. It improves access to education, healthcare, and markets.
- iii. Encourages environmentally friendly road construction.
- iv. Reduces erosion and landslide risks.
- v. Increases community participation and ownership.
- vi. Enhances long-term sustainability of road projects.
- vii. Helps reduce poverty and regional inequality.

Cons of transport plus concept:

- i. Higher initial project cost than conventional road construction.
- ii. Requires coordination among multiple government agencies and stakeholders.
- iii. Longer planning and implementation period.
- iv. Needs skilled technical and social management.
- v. Community participation may delay decision-making.
- vi. Benefits may take time to become visible.
- vii. Monitoring and maintenance are more complex.

**2. Draw a typical cross section of road with all elements. Describe briefly about each element of cross section of road.**

Answer:

Highway cross-section is the section of road cut perpendicular to the centerline axis of the road. The elements of highway cross-section influence the life of the pavement as well as riding comfort and safety. It consists of many parts such as carriageway, shoulder, median strips, traffic separators, side drain, side slope, catch drains etc.

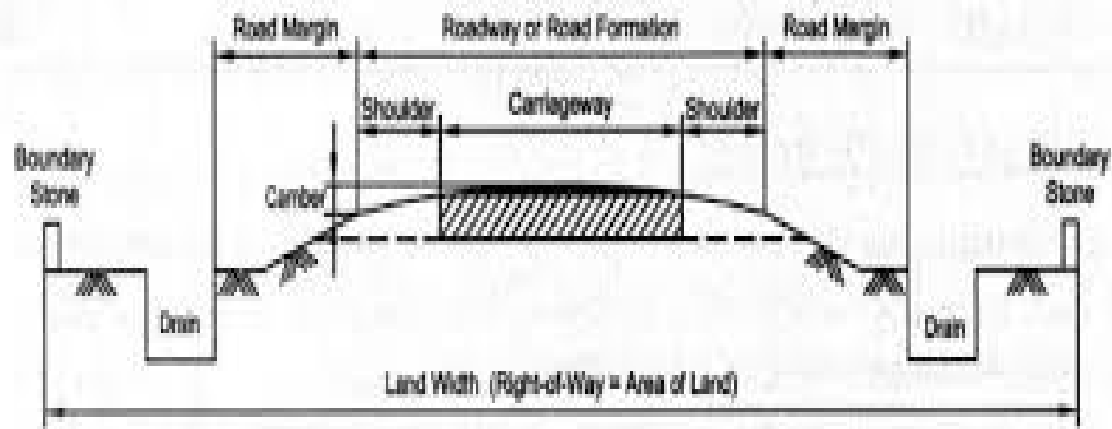


FIG. 1 CROSS-SECTION DETAILS OF A ROAD

**i. Carriageway:**

- It is the strip of road meant for vehicular traffic movement also known as pavement width.
- Width of carriageway depends upon the width of traffic lanes and number of lanes.
- According to NRS 2070, The minimum lane width for a single lane road is 3.75 m.
- Two lane road is provided with minimum of 3.5 m for each lane.

**ii. Shoulder:**

- They are the strips provided along the road edge and are intended for the accumulation of stopped vehicles.
- Shoulders serve as emergency lane for vehicles and also provide the lateral support for base and surface courses.
- According to NRS 2070, the width of shoulders on either side of carriageway shall be at least 0.75m.
- It is desirable that color and texture of shoulders be different from those of the carriageway.

**iii. Roadway width:**

- The sum of width of pavements or carriageway including separators and shoulders is called width of formation or roadway width.
- It is the top width of highway embankment or bottom width of highway cutting excluding the side drains.

**iv. Side slope:**

- According to NRS 2070, side slopes of embankment and cuttings depend upon the type of fill/ cut materials and height/depth of fill or cut.
- If natural cross slope of the ground is more than 1:5 then the ground should be cut with more than 2 m wide horizontal steps.

**v. Lay-bys, bus bays and bus stops:**

- Lay-bys are the intermittent shoulders sufficiently wide and long provided to meet the important function of shoulder where the continuous shoulder on either side cannot be provided from economic considerations.
- Bus-bays shouldn't be located too close near the intersections. It is desirable that they are located at least 40 m from the intersection on either side.

- Bus stops should be placed adjacent to the bus linear line of travel so that it doesn't need to pull over to left. It must be provided at the spacings of 200-00 m.

**vi. Right of way:**

- It is the width of land to be acquired for the road along its alignment. It must be adequate to accommodate all the cross-sectional elements of highway and reasonably provided for future development.
- According to NRS2070,
- ROW requirements for national highways= 50 m (25 m on either side of the road centerline)  
Feeder roads = 30 m (15 m on either side of road centerline)  
District roads= 20 m (10 m on either side of road centerline)

**3. Differentiate the specific planning and design considerations for the suspension bridge from those for the suspended trail bridge.**

Answer:

The rugged and complex mountainous terrain of Nepal makes suspension bridges and suspended trail bridges an ideal solution for crossing rivers and deep valleys. The longest suspension bridge in Nepal is the Susta Suspension Bridge in West Nawalparasi, measuring 1,571 meters long and 1.6 meters wide.

The differences between suspension bridge and suspended trail bridge in design considerations and specific planning are listed below:

| Suspension bridge                                                                                                       | Suspended trail bridge                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| The bridge is designed to carry motor vehicles and heavy traffic.                                                       | This bridge is designed to carry pedestrian, livestock and cyclists.                                              |
| They are of longer spans 100-200 m or more.                                                                             | They are of short to medium spans 20-300 m.                                                                       |
| Suspension bridge has wide deck of width 3.5-10 m or more for one or more traffic lanes.                                | Suspended trail bridge has narrow deck of 1-2 m suitable for pedestrian movement.                                 |
| The design load considered in suspension bridges are heavy vehicular loads, impact loads, wind loads and seismic loads. | The design load considered in suspended trail bridge are pedestrian load, livestock load, wind and seismic loads. |
| The bridge requires strong and deep foundations.                                                                        | The bridge requires comparatively lighter foundations and anchorages.                                             |
| The deck has high stiffness that minimizes vibration and deflection.                                                    | The deck has moderate stiffness and limited deck movement.                                                        |
| The construction cost of suspension bridge is very high due to heavy structural components.                             | The construction cost of suspended trail bridge is comparatively low and economical.                              |





4. **List the various impurities present in surface water sources. What are their effects on human health? Describe the various components of water treatment plant generally used for purification of water from such sources.**

Answer:

The unwanted substances found in the water that changes its physical, chemical and biological composition is known as impurities. The various impurities present in surface water sources are listed below:

i. Physical impurities:

The impurities that affect the physical composition of water such as color, odor, taste etc. are known as physical impurities.

- Suspended solid such as sand, silt and clay
- Floating debris
- Organic matter, industrial wastages

ii. Chemical impurities:

The impurities that affect the chemical composition of water such as pH, solids, salts of minerals, alkalinity etc. are known as chemical impurities.

- Dissolved salts
- Iron and manganese
- Fluoride, arsenic, nitrates, lead etc.

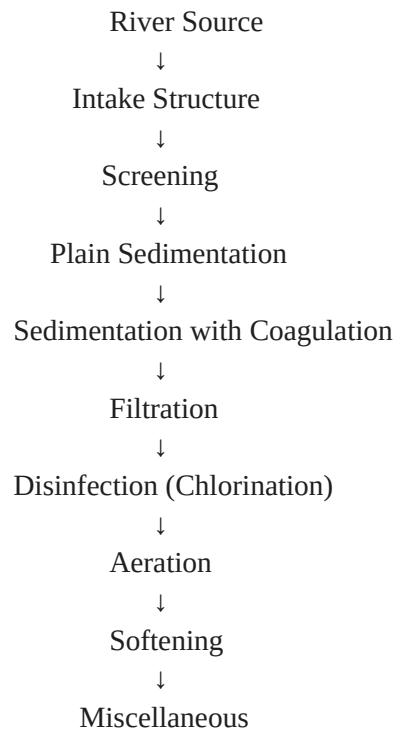
iii. Biological impurities:

the impurities that affect the biological composition of water such as pathogenic, non-pathogenic microorganisms etc.

- Bacteria
- Viruses
- Protozoa
- Helminths
- Algae

Water treatment is the process of removing physical, chemical, and biological impurities from raw water to make it safe and suitable for drinking and other domestic uses. Sundarjal water treatment plant in Kathmandu valley is one of the major water treatment plants in Nepal. Almost all major towns with piped supply (e.g., Dhulikhel, Pokhara, Bharatpur, Butwal, Nepalgunj) have smaller conventional WTPs, serving surface-water schemes.

**Schematic flow diagram illustrating the possible treatment units involved in the water treatment process:**



**Intake structures:**

- Hydraulic device constructed at or near river source to draw the required amount of water from the river source.
- Intake structures must be provided with the gate system to prevent entry of floating debris, large suspended sediments to the treatment plant.

**Screening:**

- Screening is the process of passing water through screens to remove large floating and suspended matter such as sticks, branches of tree, dead animals etc.
- Depending upon the size of the screens, there are three types of screens: coarse screen, fine screen and medium screen.
- Screening reduces the load on the treatment plant and increases the efficiency of treatment.

**Plain sedimentation:**

- The process of removal of coarse suspended particles presents in the water such as sand, silt by allowing it to remain quiescent in sedimentation tank so that the suspended material settle down at the bottom by the action of gravity is known as plain sedimentation.
- Detention period in a plain sedimentation tank= 4-8 hours.
- SOR of a plain sedimentation tank=15 to 30 m<sup>3</sup>/m<sup>2</sup>/day

**Sedimentation with coagulation:**

- The fine suspended and colloidal materials like clay particles which cannot be removed by plain sedimentation or takes long time to removal by plain sedimentation can be removed by sedimentation with coagulation.
- Coagulants such as aluminum sulphate, sodium aluminate, iron salts are used.
- Detention period for sedimentation with coagulation = 1-4 hours
- SOR for sedimentation with coagulation= 30 to 40 m<sup>3</sup>/m<sup>2</sup>/day

**Filtration:**

- Filtration is the process of passing water through the beds of sand and granular material.

- For the removal of those impurities whose specific gravity is less than or equal to that of water for which sedimentation process does not work, in such case filtration is used.
- Slow sand filter, Rapid sand filter, pressure filters are some of the common filters used for the filtration of water in treatment plants.

#### **Disinfection:**

- The process of killing pathogenic bacteria and microorganisms present in water for the safe use is known as disinfection.
- The chemicals or substances used to kill microorganisms are disinfectants.
- Chlorination is one of the most common processes of disinfecting water.

#### **Aeration:**

- The process of bringing water in contact with air to remove dissolved gasses present in water is known as aeration.
- Aeration removes CO<sub>2</sub> gasses up to 70% due to which corrosion corrosiveness of water is reduced.

#### **Softening:**

- The process of removal or reduction of temporary or permanent hardness of water is known as softening.
- Softening helps to improve the taste of water, reduce consumption of soap during washing, bathing and also reduces formation of scales in boiler.

#### **Miscellaneous:**

- It includes:
  - Removal of iron and manganese
  - Removal of color, odor, taste from the water
  - Removal of arsenic and so on.

The treatment units are arranged in such a way that it progressively remove large debris, suspended solids, colloidal particles, microorganisms, and pathogens, to ensure that the water meets drinking water quality standards before distribution to consumers.

### **5. Describe the causes of various water borne diseases transmitted into human body with their resources of transmission routes and preventive measures.**

Answer:

- Waterborne diseases are those diseases caused due to consumption of water containing pathogenic microorganisms or chemical impurities.
- Water borne diseases are also known as water quality diseases.
- These diseases commonly occur using contaminated water during bathing, washing, drinking, preparation of food or consumption of food.
- Water-borne diseases have been reported to account for about 15% of all illnesses and 8% of total deaths in Nepal, with a much higher burden among children under 5: 41% of illnesses and 32% of deaths.
- The statistics also states that 3,500 children die each year in Nepal due to water-borne diseases.
- The different water borne diseases transmitted through polluted water are listed below:
  - ✓ Typhoid (by Salmonella Typhi)
  - ✓ Cholera (by Vibrio Cholera bacteria)
  - ✓ Dysentery (by Shigella bacteria)
  - ✓ Paratyphoid (by Salmonella Para typhi)
- The causes of various water borne diseases transmitted by human body are explained below:
  - Contaminated drinking water:
    - Drinking water contaminated with human and animal feces are the major cause of water borne disease, when such water directly or indirectly reaches on the stomach of person.

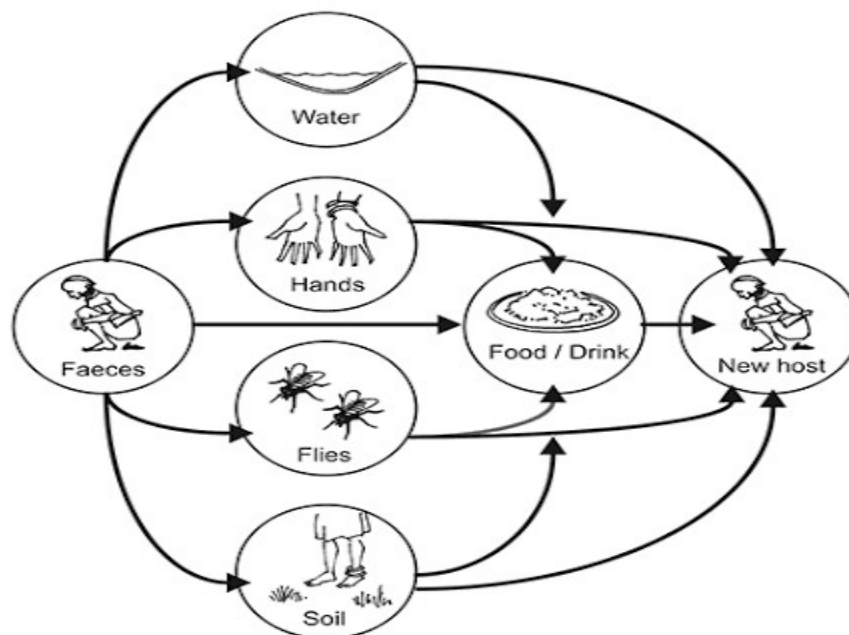


- Contamination of water may occur due to open defecation nearby the water sources, disposal of untreated sewage directly into rivers and so on.
- ii. Poor sanitation:
  - Poor sanitary conditions in the household or community leads to contamination of water sources which leads to transmission of various water borne diseases.
  - Unhealthy sanitary habits such as washing clothes in rivers, ponds, open defecation, lack of toilets etc. results in pollution of water sources.
- iii. Poor personal hygiene:
  - Lack of proper personal hygiene such as not washing hands after using toilets or before eating foods, washing foods with contaminated water etc. causes the bacteria to reach inside the body of human causing water-borne diseases.
- iv. Inadequate treatment of drinking water:
  - Drinking un-filtered and un-treated water often leaves the bacteria in it, which reaches to the human body causing water borne diseases.
- v. Environmental pollution:
  - Residential sewage, industrial wastages, and animal waste pollutes the water resources and increase the disease risk.

The water-borne diseases transmit through a route which is known as faecal oral transmission route.

#### **Faecal Oral Transmission:**

- Organisms from the excreta of animals reaches to water sources directly or sewage disposal.
- Then this water is transmitted in the human body orally from the food or contaminated water
- This process is known as faecal oral transmission.
- Water washed and water borne diseases is caused by faecal oral route.
- Transmission of the water-related disease is related to water supply and sanitation due to inadequate and improper disposal of waste and unsafe drinking water.
- The diagram of faecal oral transmission is shown below:



#### **➤ Preventive measures:**

- i. Proper washing of hands after the use of toilet and before meal.
- ii. Proper covering of food and water during storage.
- iii. Water must be fetched from clean water sources.

- iv. Sewage should not be disposed at or near the water resources.
- v. Only properly filtered water must be drunk.
- vi. Avoid fly nuisance by proper disposal of animals and organic wastes.
- vii. Public awareness and improving the living habits of people.

6. **Mention the points to be considered topographically, geologically, hydrologically and economically at the planning, layout and design stage of a small hydropower project.**

Answer:

Planning and designing a small hydropower project require detailed investigations of the site to ensure the project is technically feasible, economically viable, environmentally sustainable, and safe. The major considerations are as follows:

- i. Topographical considerations:
  - Availability of sufficient head between intake and powerhouse.
  - Proper location of the intake, desanding basin, canal/penstock, and powerhouse.
  - Stable and accessible terrain for construction.
  - Short and economical alignment of canal or penstock.
  - Availability of adequate land for future expansion.
- ii. Geological considerations:
  - Geological stability of dam, intake and powerhouse sites.
  - Slope stability considerations
  - Seepage and permeability of foundation materials
  - Earthquake risk and tectonic activities
  - Availability of construction materials near the site
  - Presence of faults, folds, cracks, joints, fractures etc.
- iii. Hydrological considerations:
  - Availability of adequate discharge in the river throughout the year
  - Rainfall and catchment characteristics
  - Flood estimation and design flood
  - Sediment load and silt characteristics
  - Climate change impacts on river flow
  - Water requirements for irrigation, drinking water and environmental flow.
- iv. Economic considerations:
  - Initial construction cost
  - Operation and maintenance cost
  - Cost of acquisition and compensation
  - Benefit cost ratio
  - Creating local employment opportunities and socio-economic benefits.

7. **What is uplift? How can uplift be determined? Show this with a neat sketch of the cross-section of a weir with two rows of sheer piles at the ends. Explain how will determine the uplift pressure at any points of the weir foundation.**

Answer:

The upward hydrostatic pressure exerted by seepage water beneath the floor of a hydraulic structure such as a weir, barrage, or dam is known as uplift. Water seeps from the upstream side to the downstream side through the permeable foundation, producing pressure under the floor. If this pressure is not properly resisted, it may cause cracking of the floor, piping failure, sliding and overturning of the structures etc.

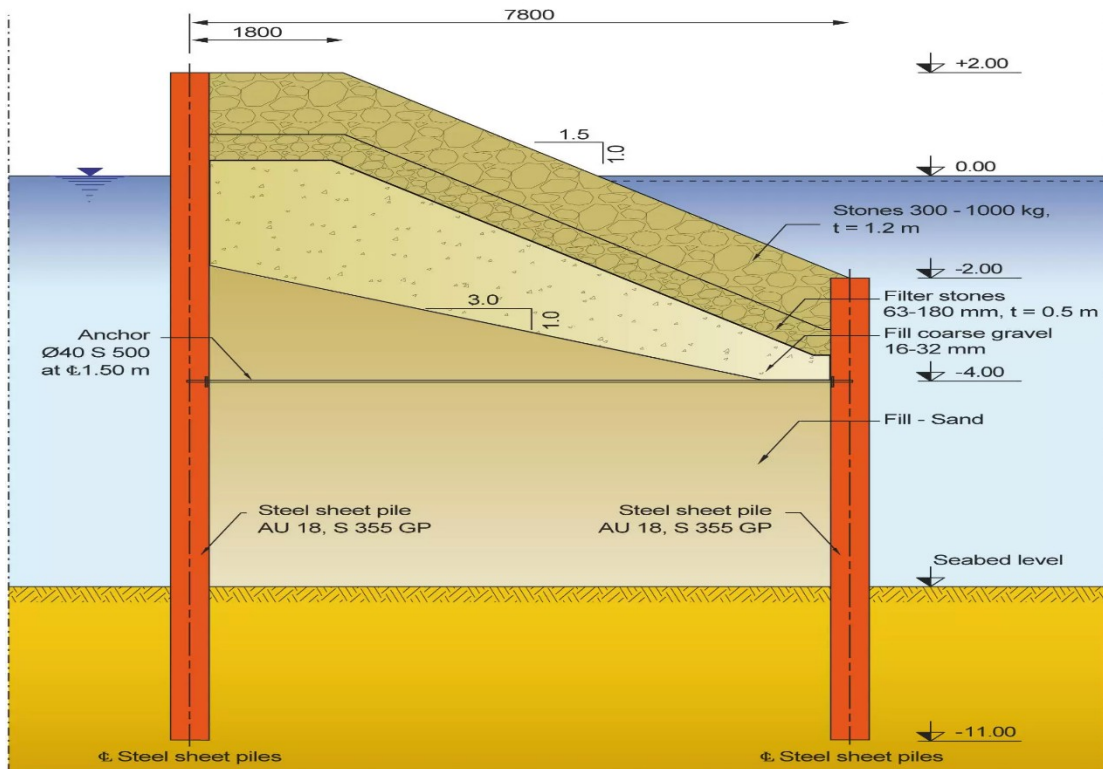
The uplift pressure is determined by studying the seepage of water beneath the weir floor. The commonly used methods are:

- i. Bligh's Creep Theory
- ii. Lane's Weighted Creep Theory
- iii. Khosla's Theory of Independent Variables

Khosla's theory is mostly preferred as it is more advanced theory and considers depth of sheet piles, floor length, hydraulic gradient and pressure distribution beneath the floor. The uplift pressure at any point is obtained from the percentage pressure head calculated using Khosla's curves.



The cross-section of a weir with two rows of sheet piles are as follows:



Upward hydrostatic pressure exerted by seepage water beneath the floor of a hydraulic structure is known as uplift pressure. To ensure the safety of the floor of hydraulic structure against uplift and piping, uplift pressure must be determined.

Design procedure for determining uplift pressures in a floor of hydraulic structures:

- Upstream and downstream high flood level or water level is noted to calculate the total hydraulic head generating seepage.
- Structural data such as length of the floor, depth and location of upstream and downstream cutoff, thickness of floor, foundation of soil properties is collected.
- Residual head is determined at different points under the floor using Khosla's theory, Bligh's theory, Lane's weighted Creep theory.
- Uplift pressure is calculated.

At any point,

$$p = \gamma_w \times h$$

where:

$$p = \text{uplift pressure (kN/m}^2\text{)}$$

$$\gamma_w = \text{unit weight of water } (\approx 9.81 \text{ kN/m}^3 \vee 10 \text{ kN/m}^3)$$

$$h = \text{residual pressure head (m)}$$

- Uplift pressure diagram is plotted beneath the floor. The maximum pressure occurs near the upstream end and decreases towards the downstream end.
- Floor thickness of structure should be determined. It should be enough to resist uplift.

Check:

$$\text{Weight of floor} \geq \text{Uplift force} \times \text{Factor of Safety}$$

A factor of safety of about 1.2–1.5 is commonly adopted.

- vii. Increase the floor thickness if necessary. Provide upstream and downstream cutoffs. Install drainage galleries or pressure relief holes where applicable.

8. **Describe about the river training work. Give classification and types of river training work in Nepal.**

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

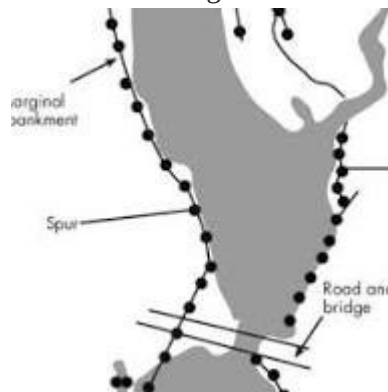
The objectives of river training works are as follows:

- i. To protect river banks from erosion by preventing scouring and collapse of river banks.
- ii. To protect hydraulic structures and ensures safety of bridges, dams, weirs and other structures from erosion and scour.
- iii. To provide safe waterway for navigation by maintaining sufficient depth and stale channel.
- iv. To reduce flood damage by regulating the river flow.
- v. To minimize silt accumulation near bridges, intakes etc.

There are different methods of river training works. Some of them are described below:

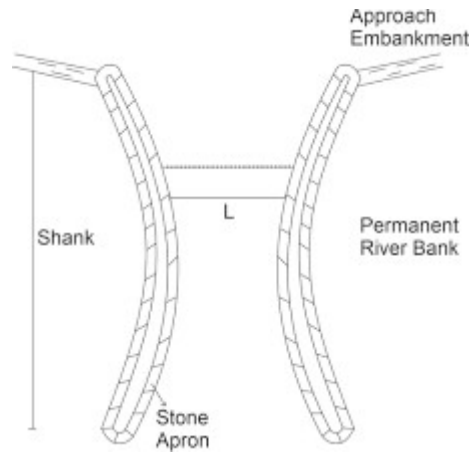
i. **Marginal embankments/ levees:**

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



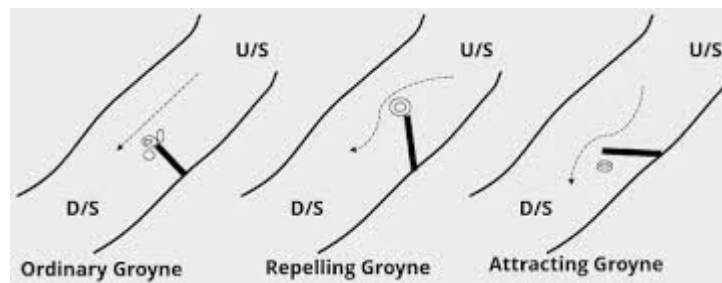
ii. **Guide banks:**

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



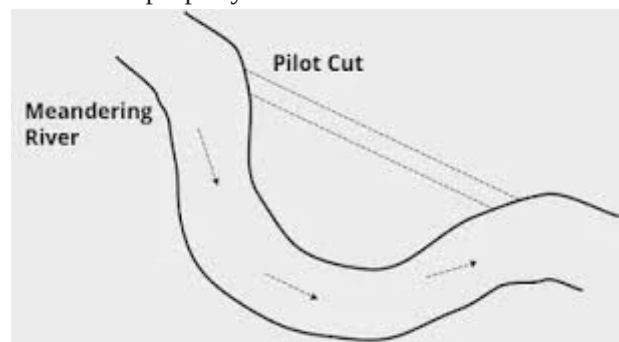
### iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



### iv. Artificial cutoff:

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.



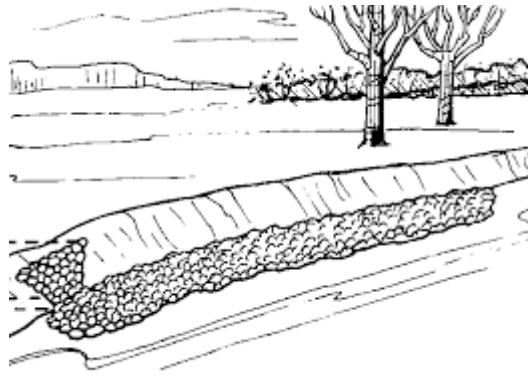
### v. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



**vi. Pitched islands:**

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



**vii. Bandalling:**

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.



**9. Write short notes on:**

**a. Indigenous technology in building construction**

**b. Urban planning needs in Nepal**

Answer:

**a. Indigenous technology in building construction**

- Indigenous technology in building construction refers to the use of traditional, locally developed construction techniques, local materials, and skills that have evolved over generations to suit the local climate, geography, culture, and available resources.
- These methods are generally economical, environmentally friendly, and sustainable.
- Some of the common examples of indigenous technology in building construction is Stone masonry with mud mortar in hill regions, traditional Newari brick and timber houses in the Kathmandu Valley, slate-roofed stone houses in mountainous districts.
- Some of the common indigenous building technologies are:

- i. Mud construction:
  - ✓ Loc cost construction in which walls are made up of sun-dried mud bricks or compacted earth.
  - ✓ It provides good thermal insulation.
- ii. Stone masonry:
  - ✓ It is made up of locally available stones.
  - ✓ It is commonly used in hilly and mountainous regions of Nepal.
- iii. Bamboo construction:
  - ✓ It is lightweight, renewable and economic construction.
  - ✓ It is used for walls, roofs, scaffolding and temporary structures.
- iv. Thach roof construction:
  - ✓ In this type of construction, roofs are made up of straw, reeds or grasses.
  - ✓ It provides excellent thermal insulation.
- v. Slate construction:
  - ✓ In this type of construction, flat stone slabs are used as roofing materials.
  - ✓ It is common in mountainous areas.

Advantages:

- i. Uses locally available materials, reducing transportation costs.
- ii. Low construction cost and promotes local employment.
- iii. Environmentally friendly and sustainable construction.
- iv. Provides good thermal comfort suited to local climate.

Disadvantages:

- i. Lower strength and durability compared to modern RCC buildings.
- ii. Vulnerable to earthquakes if not properly reinforced.
- iii. Requires regular maintenance.
- iv. Limited height and span of structures.
- v. Quality depends on the skill of local craftsmen.

b. Urban planning is the process of planning, designing, and managing the physical, social, economic, and environmental development of towns and cities to ensure orderly, sustainable, and livable urban growth.

Urban planning is important in Nepal due to rapid urbanization, population growth, rural-to-urban migration, and increasing demand for housing, transportation, water supply, sanitation, and other urban services. Effective urban planning helps improve the quality of life while protecting natural resources and reducing disaster risks.

**Urban planning is important in Nepal because of the following reasons:**

- i. Urban planning promotes planned urban **growth** and prevents unplanned settlements.
- ii. It provides adequate infrastructure, such as roads, water supply, sewerage, drainage, electricity, and solid waste management.
- iii. It improves transportation systems and reduces traffic congestion.
- iv. Urban planning promotes proper land **use** through zoning and development control.
- v. It helps to protect the environment by conserving open spaces, rivers, forests, and heritage sites.
- vi. It reduces disaster risks, especially from earthquakes, floods, and landslides.
- vii. It also improves public health through better sanitation and waste management.
- viii. It supports economic development by creating efficient commercial and industrial areas.

**10. a. Explain the impacts of modern construction technology in the society.**

**b. Explain the participatory approach in local infrastructure construction at various stages of construction.**

Answer:

a. Modern construction technology involves the use of advanced materials, machinery, digital tools, and innovative construction methods to improve the planning, design, and execution of construction projects. The positive impacts of modern construction technology in the society are as follows:

- i. Improved construction speed:  
The project completion time can be greatly reduced by the use of modern equipment, prefabrication and modular construction.
- ii. High quality and durability:  
Use of advanced materials and quality control techniques helps to produce stronger, safer and long-lasting structures.
- iii. Enhanced safety:  
Modern machinery, protective equipment and safety management systems reduce workplace accidents.
- iv. Cost efficient:  
Efficient use of materials, equipment and labor in construction work reduces the construction and maintenance cost over time.
- v. Environmental sustainability:  
Green buildings, energy-efficient designs, recycled materials and renewable energy technologies reduce environmental impacts.

The negative impacts of modern construction technologies are listed below:

- i. High initial investment:  
Advanced machinery and technologies require significant capital investment.
- ii. Environmental concern:  
Excessive use of cement, steel and heavy equipment may increase carbon emissions and resource capitalization.
- iii. Construction waste:  
Large projects generate significant amount of demolition and construction waste if not managed properly.

b. A method of planning, design, construction, operation, and maintenance of local infrastructure in which the local community actively participates in decision-making, implementation, monitoring, and maintenance is known as participatory approach. It ensures that projects meet local needs, promote ownership, and improve sustainability. In Nepal, Local Self Governance Act (1999) laid early foundational rules for engaging citizens in local bodies.

The participatory approach in local infrastructure construction at various stages of construction are given below:

- i. Planning stage:
  - Community needs and priorities are identified through public meetings and consultations.
  - Project location with community consensus is selected.
  - Social, environmental and economic assessments are conducted.
  - Former user committees and stakeholder roles are defined.



- ii. Design stage:
  - Local people are consulted regarding design alternatives.
  - Indigenous knowledge must be incorporated with the use of locally available materials wherever appropriate.
  - The design must meet the local requirements and environmental conditions.
  - Project cost, benefits and implementation schedules must be discussed with stakeholders.
- iii. Construction stage:
  - Local labor must be mobilized and community contribution must be encouraged.
  - Transparency must be maintained in procurement stage, budgeting and quality control.
  - Disputes must be resolved through community participation.
- iv. Monitoring and evaluation stage:
  - Regular inspections must be conducted with engineers and local representatives.
  - Community members must also monitor construction progress and quality.
  - Transparency and accountability must be maintained through public reporting.
- v. Operation and maintenance stage:
  - Operation and maintenance committees are established.
  - Regular repairs are carried out to ensure long-term functionality.